Service Bulletin

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Turbocharger Lubricating Oil Leak Troubleshooting	

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Issue

Lubricating oil leaks on the compressor or turbine side of turbocharger are most often the result of an upstream engine component and/or system malfunction and very rarely indicate a turbocharger malfunction. If lubricating oil leaks without axial or radial play issues are observed, do **not** replace the turbocharger for oil leak related issues. This document will explain the symptoms, root cause, and resolution for turbocharger lubricating oil leaks and provide a technical explanation of what causes turbocharger lubricating oil leaks.

Symptom

- Turbocharger compressor side lubricating oil leak (CSOL)
- Turbocharger turbine side lubricating oil leak (TSOL)
- Lubricating oil staining or pooling in bottom of turbocharger compressor housing or lubricating oil found in air intake piping downstream from turbocharger
- Apparent leaks from the turbocharger compressor housing or charge air cooler plumbing

Root Cause

- Root cause of problem can be related to:
  - Extended idle time
  - Restricted air cleaner
  - Low boost pressure (caused by transient operating conditions, boost leaks from intake piping or charge air cooler)
  - Elevated crankcase pressure caused by:
    - Contaminated lubricating oil
    - Overfilled lubricating oil
    - Cylinder or cylinder head malfunction
    - Restricted crankcase ventilation system
  - Restricted turbocharger drain line
- If any of the conditions listed above are present, an incorrect pressure differential across the turbocharger seal(s) may occur resulting in apparent lubricating oil leakage into the turbocharger compressor or turbine housings. Turbocharger seal functionality can be restored by correcting any identified root cause.
- An under-performing or restricted closed crankcase ventilation system can allow engine lubricating oil from the lubricating oil pan to be pulled into the compressor housing of the turbocharger then pushed through the charge air cooler.

- An under-performing or restricted closed crankcase ventilation system includes but **not** limited to:
  - Obstructed closed crankcase ventilation system oil drain
  - Malfunctioned open closed crankcase ventilation system oil drain check valve – oil is drawn up the closed crankcase ventilation system oil drain and into the turbocharger compressor housing
  - Clogged closed crankcase ventilation system filter element

Resolution

- Do **not** replace turbocharger if **only** symptom is lubricating oil found in turbocharger compressor or turbine housing or in intake/exhaust piping. Lubricating oil leaks from compressor (cold side) or turbine (hot side) seals do **not** indicate a malfunction of turbocharger oil seal.
- When troubleshooting suspected Turbocharger compressor seal lubricating oil leak, **never** attempt to replicate the lubricating oil leak by operating the engine with the turbocharger compressor outlet piping disconnected as this will always cause a turbocharger compressor seal lubricating oil leak regardless of turbocharger health or age.
- Reference updated Turbocharger Service Manual Procedure and diagnostics when a CSOL or TSOL situation is encountered.
- If turbocharger radial and axial end play is still in specification, clean and install turbocharger and charge air cooler inlet pipe.
- After repair is complete and the engine is operated, residual lubricating oil could be pushed from the turbocharger compressor housing and produce lubricating oil streaks in or on the turbocharger piping. This does not constitute a malfunction.
- If troubleshooting is completed and root cause of failure is **not** identified, contact Cummins Care at 1-800-CUMMINS™ for troubleshooting direction. Replace the turbocharger **only** if directed by Cummins Care™.

Technical Summary

Lubricating oil found in turbocharger compressor housing, turbine housing, charge air piping, charge air cooler, and/or exhaust piping is related to the piston seal design used in the turbocharger. Cummins Inc. turbocharger oil seals are a metal piston-ring seal (See Figure 1 below) that is designed to function under a positive pressure differential between turbine or compressor housing pressure (related to exhaust or boost pressure) and turbocharger bearing housing pressure (related to crankcase pressure). During certain duty cycles, such as idling and accelerations, the turbocharger turbine or compressor housing pressure drops below bearing housing pressure, forcing lubricating oil past turbocharger seal and into turbocharger turbine or compressor housing. See Figure 2 below.

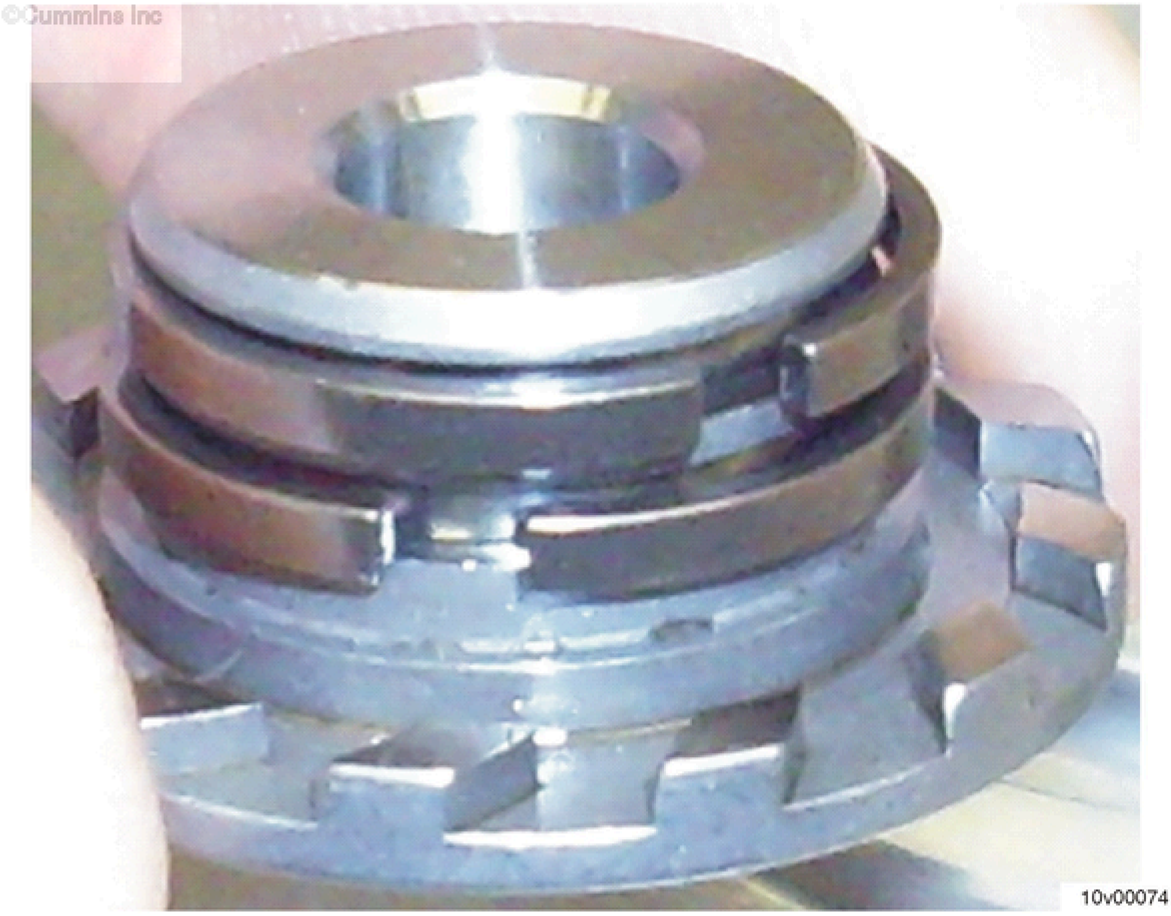

Figure 1, Example of Turbocharger Compressor Oil Seal.

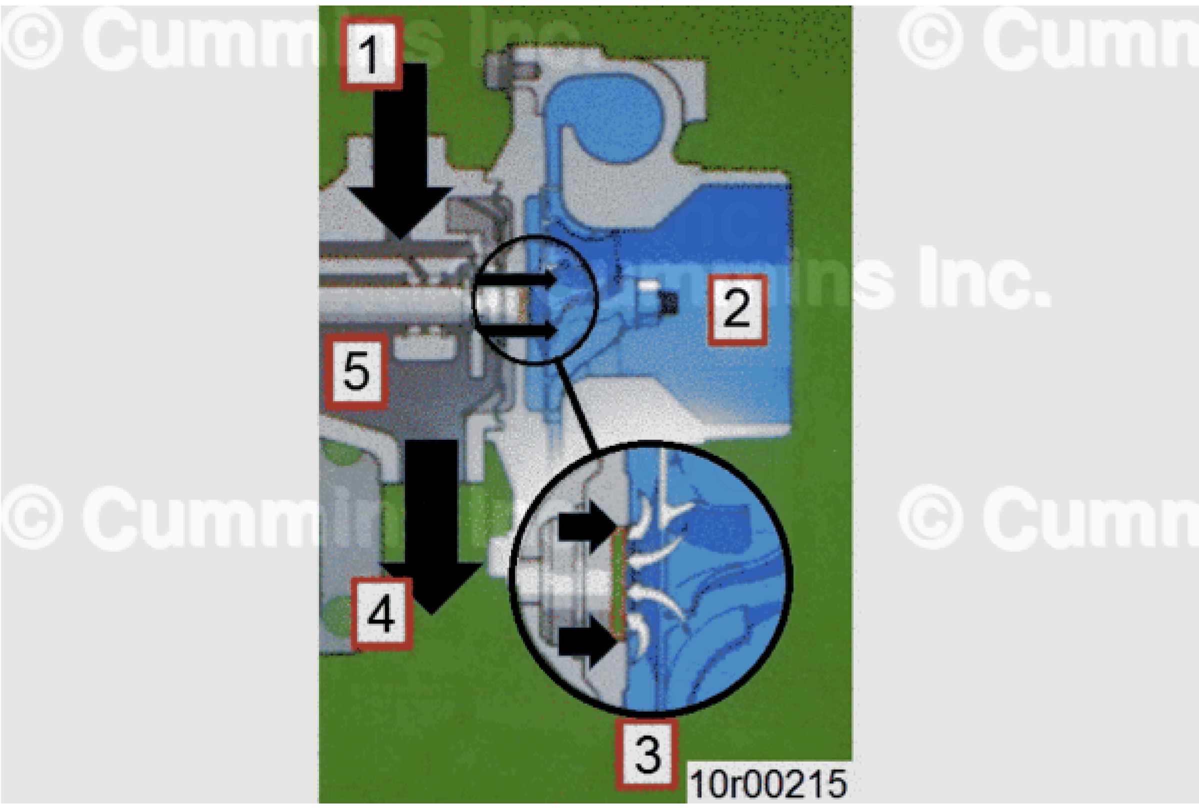

Figure 2, Turbocharger Oil Leak Diagram.

1. Turbocharger Lubricating Oil Supply Line

2. Intake Air Pressure
3. Lubricating Oil and Air Pressure
4. Turbocharger Lubricating Oil Drain Line
5. Crankcase Pressure

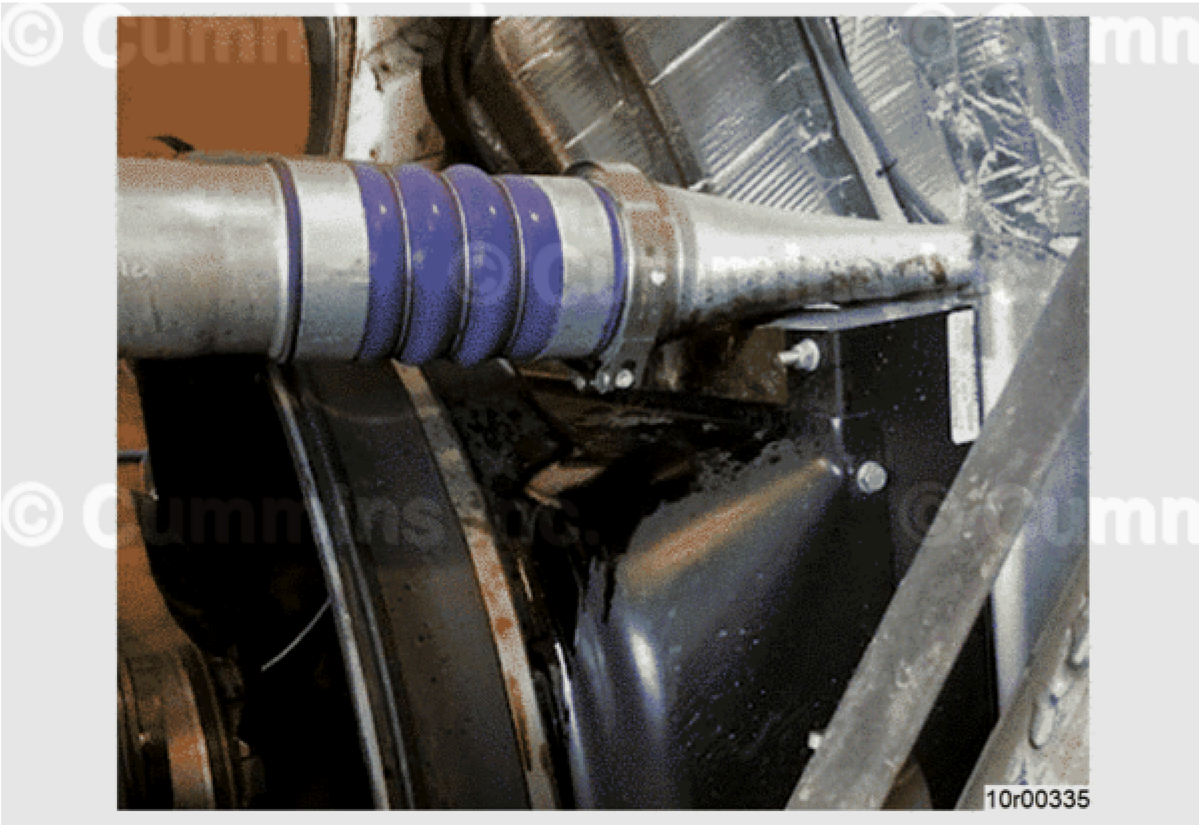

Figure 3, Charge Air Cooler Plumbing Leaking Oil.

If the charge air cooler plumbing is leaking oil, see Figure 4 below for example of oil in compressor inlet due to closed crankcase ventilation oil carryover.

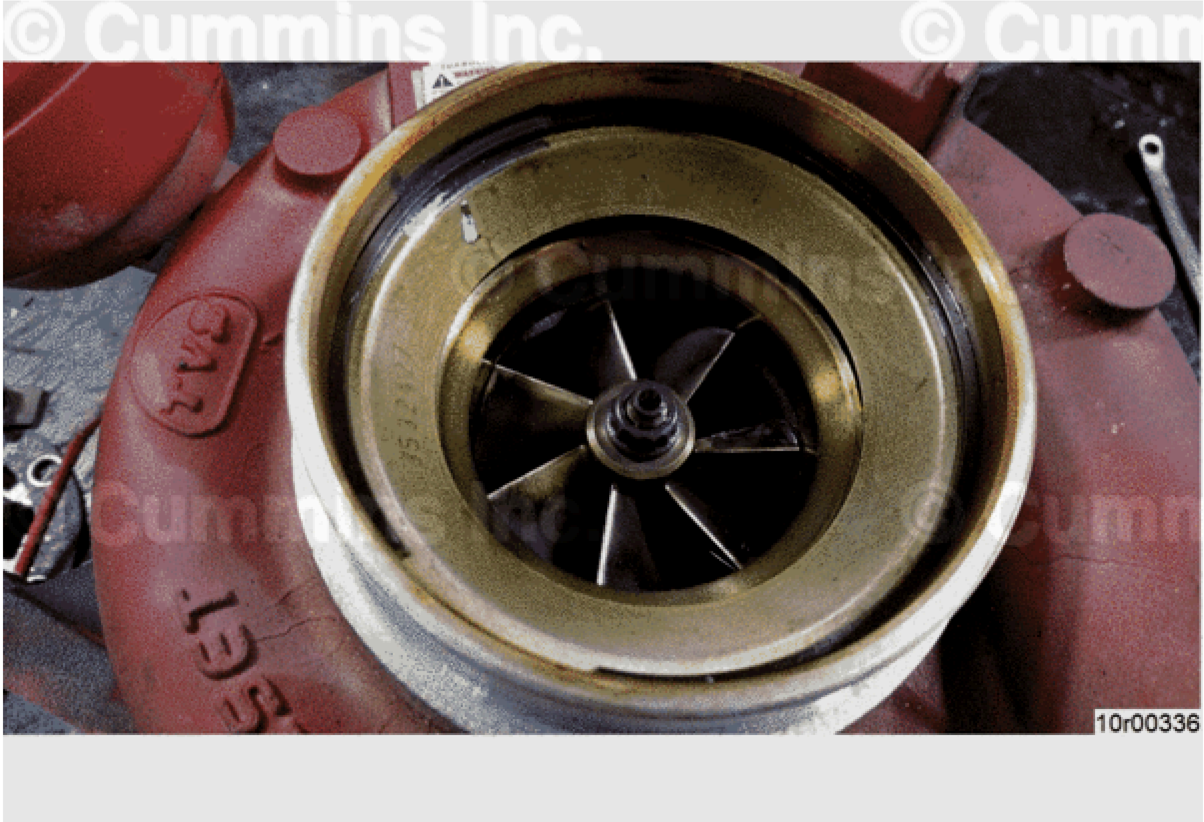

Figure 4, Turbocharger Compressor Inlet (Acceptable with Oil Deposits From CCV Oil Carryover).

If there is evidence of oil in the turbocharger compressor inlet, check the closed crankcase ventilation system. See Service Bulletin, Closed Crankcase Ventilation System Engine Lubricating Oil Carryover and Associated Turbocharger Lubricating Oil Leak, Bulletin 5659915.

Document History

Date	Details
2017-11-10	Module Created
2017-11-16	Updated Figure 2 captions.
2018-8-28	none
2019-2-22	none
2020-11-10	Added more information about leaks.
2024-4-5	Adding additional comment: Replace the turbocharger **only** if directed by Cummins Care™.

Last Modified: 05-Apr-2024